

The safety of allogeneic innate lymphocyte therapy for glioma patients with prior cranial irradiation

Larisa Pereboeva · Lualhati Harkins · Shun Wong ·
Lawrence S. Lamb

Received: 23 April 2014 / Accepted: 24 January 2015 / Published online: 13 February 2015
© Springer-Verlag Berlin Heidelberg 2015

Abstract The standard treatment of high-grade glioma presents a combination of radiotherapy, chemotherapy and surgery. Immunotherapy is proposed as a potential adjunct to standard cytotoxic regimens to target remaining microscopic disease following resection. We have shown *ex vivo* expanded/activated $\gamma\delta$ T cells to be a promising innate lymphocyte therapy based on their recognition of stress antigens expressed on gliomas. However, successful integration of $\gamma\delta$ T cell therapy protocols requires understanding the efficacy and safety of adoptively transferred immune cells in the post-treatment environment. The unique features of $\gamma\delta$ T cell product and the environment (hypoxia, inflammation) can affect levels of expression of key cell receptors and secreted factors and either promote or hinder the feasibility of $\gamma\delta$ T cell therapy. We investigated the potential

for the $\gamma\delta$ T cells to injure normal brain tissue that may have been stressed by treatment. We evaluated $\gamma\delta$ T cell toxicity by assessing actual and correlative toxicity indicators in several available models including: (1) expression of stress markers on normal primary human astrocytes (as surrogate for brain parenchyma) after irradiation and temozolomide treatment, (2) cytotoxicity of $\gamma\delta$ T cells on normal and irradiated primary astrocytes, (3) microglial activation and expression of stress-induced ligands in mouse brain after whole-brain irradiation and (4) expression of stress-induced markers on human brain tumors and on normal brain tissue. The lack of expression of stress-induced ligands in all tested models suggests that $\gamma\delta$ T cell therapy is safe for brain tumor patients who undergo standard cytotoxic therapies.

L. Pereboeva (✉)
Division of Hematology and Oncology, School of Medicine,
University of Alabama at Birmingham (UAB), 1824 6th Ave S,
WTI 510C, Birmingham, AL 35294, USA
e-mail: lpereb@uab.edu

L. Harkins
Division of Hematology and Oncology, School of Medicine,
University of Alabama at Birmingham (UAB), 1824 6th Ave S,
WTI 530, Birmingham, AL 35294, USA
e-mail: customimmuno8leh@hotmail.com

S. Wong
Department of Radiation Oncology, Sunnybrook Health Sciences
Centre, University of Toronto, 2075 Bayview Ave, Toronto, ON
M4N 3M5, Canada
e-mail: Shun.Wong@sunnybrook.ca

L. S. Lamb
Division of Hematology and Oncology, School of Medicine,
University of Alabama at Birmingham (UAB), 1824 6th Ave S,
WTI 510F, Birmingham, AL 35294, USA
e-mail: ls1amb@uabmc.edu

Keywords Gamma delta T cells · Glioblastoma · Cancer immunotherapy · Irradiation · Safety

Abbreviations

BrdU	Bromodeoxyuridine
CAR	Chimeric antigen receptor
CTL	Cytotoxic T lymphocytes
DAPI	4',6-diamidino-2-phenylindole, nuclear counter-staining dye
DDR	DNA damage response
E:T	Effector-to-target ratio
FFPE	Formalin-fixed paraffin-embedded specimens
GBM	Glioblastoma multiforme
$\gamma\delta$ T	Gamma delta T cells
IHC	Immunohistochemistry
IL	Interleukin
LAK	Lymphokine-activated killer cells
NK	Natural killer cells
TCR	T cell receptor

TMZ	Temozolomide
TNF α	Tumor necrosis factor α
UAB	University of Alabama at Birmingham
ZOL	Zoledronic acid

Introduction

The standard treatment of malignant high-grade glioma includes a combination of radiotherapy, chemotherapy and surgical techniques. Despite the latest advances in all three modalities, the median survival for glioma patients remains a dismal 12 months and disease remains virtually incurable. To advance glioma treatment, cellular therapies employing various types of immune cells have advanced to the stage of clinical trials in the context of glioblastoma multiforme (GBM). These include lymphokine-activated killer cells (LAK), natural killer cells (NK), cytotoxic T lymphocytes (CTL) and chimeric antigen receptor (CAR)-grafted T cells [1–3]. However, the most relevant and efficient cell type for the treatment of GBM is yet to be determined.

Among immune cell types tested, $\gamma\delta$ T cells have been recognized as a subset with clear benefits for cell therapy of solid tumors [3]. Circulating $\gamma\delta$ T cells are a minor T cell population (1–10 % of peripheral blood T cells) that manifests properties of innate immunity with some adaptive functions. Unique features of $\gamma\delta$ T cell are related to their biology and in general can be described as host homeostasis or stress surveillance [4]. To this end, $\gamma\delta$ T cells are equipped with the molecular armamentarium associated with clonally distributed TCRs and effector capabilities similar to $\alpha\beta$ T cells. However, unlike $\alpha\beta$ T cells, $\gamma\delta$ T cells do not require peptide recognition and priming and are not constrained by the selectivity and restriction of the MHC. At the same time, $\gamma\delta$ T cells display a range of innate pleiotropic effector functions including rapid secretion of chemokines and cytokines, as well as target cell lysis. Importantly, $\gamma\delta$ T cells recognize stress-induced NKG2D ligands often present on tumor cells. This recognition, both via NKG2D and the T cell receptor (TCR), is a significant functional component in both their natural biological function and in the potential anti-tumor therapeutic use of adoptively transferred $\gamma\delta$ T cells. Indeed, these unique biological properties together with potent cytotoxic anti-tumor activity have stimulated application of $\gamma\delta$ T cell-based immunotherapies in a variety of cancers [5, 6]. However, the same biological properties may present a basis for unwanted toxicities, thus warranting a rigorous preclinical safety testing.

Our previous findings have suggested that $\gamma\delta$ T cell immunotherapy is effective as a single modality in cases where only localized microscopic disease is present. On this

basis, $\gamma\delta$ T cell immunotherapy may have greater therapeutic potential when employed as adjunct to standard cytotoxic therapies applied after surgical debulking in combination with standard cytolytic treatment such as the Stupp regimen [7]. This approach can involve intracranial injection of cells into the stressed environment, creating the potential for both efficacy against tumor and potential for injury to normal brain tissue. For instance, expression of stress molecules on tumor cells sensitizes them to recognition and lysis by $\gamma\delta$ T cells that are predominantly innate effectors. At the same time, any pre-treatment conditions may also sensitize surrounding normal tissue to all natural toxicities of cell product and potentially create new toxicities due to establishing a pro-inflammatory environment and apoptosis/death of target and bystander cells. Inflammation is considered a hallmark of irradiation [8–10], where early events include pro-inflammatory cytokine induction, such as TNF α , IL-1 and IL-6 [11, 12]. Potential $\gamma\delta$ T cell therapy-related toxicities in this environment may include direct killing of brain parenchyma cells and toxicities related to unregulated cytokine release. Thus, the possibility exists that such treatment modality compromises safety to the surrounding brain tissue. Although we have consistently shown that ex vivo expanded/activated $\gamma\delta$ T cells do not kill cultured normal human astrocytes and do not damage neural tissue when injected into syngeneic mice [13], the potential for injury under intense pre-treatment radiation therapy applied to brain tumor patients is not known. Using immunocompetent animal models, cultured astrocytes and tumor lines and archival human histologic specimens, we investigated the potential for allogeneic innate lymphocyte cell therapy comprised of ex vivo expanded/activated $\gamma\delta$ T cells and a minor population of NK cells to injure the surrounding normal brain tissue that may have been stressed by prior radiation therapy.

Methods

Reagents, cell lines, animals and tumor specimens

Recombinant human IL-2 was obtained from MACS Miltenyi Biotec Inc (Auburn, CA). Zoledronic acid (ZOL) was obtained from Novartis Oncology, East Hanover, NJ. Antibodies by flow cytometry were from R&D Systems (Minneapolis, MN) and included MICA, MICB, ULBP1, ULBP2, ULBP3 and ULBP4 and matching isotype controls. Antibody for ICH on mouse brain sections included CD3 (Abcam, Cambridge, MA) and MULT-1 (eBioscience, San Diego, CA). Primary human astrocytes and astrocyte media were obtained from ScienCell Research Labs (Carlsbad, CA).

C56Bl6 mice were used to evaluate microglial activation and expression of stress ligand in irradiated mouse brain.

The whole brain of 10-week-old mice was irradiated using a dose of 10 and 17 Gy as previously described [14].

Human brain tumor specimens (GBM-15 patients) were obtained at UAB at the time of surgical resection as approved by IRB. Formalin-fixed paraffin-embedded (FFPE) specimens of metastatic breast cancer brain tumors (three patients) were obtained from Dr. Mitchell S. Berger, Department of Neurological Surgery, University of California, San Francisco, CA.

Expansion and activation of human $\gamma\delta$ T cells

$\gamma\delta$ T cells from healthy volunteers were expanded according to a modified version of a previously described method [13]. Briefly, mononuclear cells were isolated by density gradient centrifugation and resuspended at 1×10^6 /ml in RPMI 1640 with 10 % FBS serum, 100 U/ml IL-2 and 2 μ M zoledronic acid. The culture was maintained for 14 days with addition of 50 U/ml IL-2 on post-culture days 2, 5, 7, 9 and 12 and complete media to maintain cell density. Composition, purity and viability were determined by flow cytometry at culture days 0, 7 and 14.

Flow cytometry for NKG2D ligands detection

Expression of stress-associated molecules was measured by flow cytometry in normal, irradiated and TMZ-treated human astrocytes and U251 cell line. Cells were irradiated using X-ray irradiation system (X-RAD 320, Precision X-ray Inc.; North Branford, CT) at doses of 250 and 500 cGy or treated with TMZ at concentration of 200 μ M for 4 h. Stress ligands expression was assessed 24 h post-XRT and 4 or 24 h post-TMZ treatment. Cells labeled with monoclonal antibodies to MICA, MICB, ULBP1, ULBP2, ULBP3 and ULBP4 were acquired on a FACS Canto II cytometer and analyzed using DiVa software (BD Biosciences; San Jose, CA).

Cytotoxicity assay

Cytotoxicity of expanded $\gamma\delta$ T cells to normal and irradiated astrocytes was determined in vitro by flow cytometry using PKH26 membrane dye (Sigma; St. Louis, MO) according to the manufacturer's protocol. T cells were added to the target cells at effector-to-target ratios of 0:1 (Background), 5:1, 10:1 and 20:1, following incubation for 4 h at 37 °C. ToPro-Iodide viability dye (Molecular Probes; Eugene, OR) was added prior to acquisition on the flow cytometer. Cytotoxicity was calculated as: $(\text{ToproIodide} + \text{PKH26} + \text{events}/\text{total PKH26} + \text{events}) \times 100$. Three different primary $\gamma\delta$ T cell cultures were used to evaluate cytotoxicity on primary astrocytes with or without irradiation. Data were presented as a mean percentage

of target cell lysis ($\pm$ SD) obtained in three independent experiments.

Immunohistochemistry (IHC)

Formalin-fixed paraffin-embedded (FFPE) sections of mouse brain were post-fixed in neutral buffered formalin followed by antigen retrieval with Rodent Decloker (Biocare Medical CA) for CD3 staining or AR-10 for MULT-1 staining. Sections were blocked with Rodent Blocker (Bio Care medical, Richmond CA) for 30 min at RT. Sections were incubated with CD3 (ABCAM) or MULT-1 (eBioscience) antibodies overnight at 4 °C following secondary goat anti-rabbit antibodies (Biocare Medical, Richmond CA) for CD3 or Armenian Hamster STATQ link secondary antibody (Innovex Biosciences, Richmond CA) for Multi-1 for 30 min at RT. Sections were next developed with streptavidin-labeled peroxidase and Turbo DAB (Innovex Biosciences, Richmond CA) for 2 min or till signal appeared.

Formalin-fixed paraffin-embedded (FFPE) sections of human GBM and metastatic breast cancer brain tumors were used for NKG2D ligand immunohistochemistry. Sections of human normal brain provided by the Tissue Procurement Facility of the UAB Comprehensive Cancer Center were used as the antibody staining control. Deparaffinized sections were post-fixed in 4 % neutral buffered formalin followed by antigen retrieval with Citra Plus (Biogenex Laboratories, Fremont CA). Sections were block sequentially with avidin, biotin (Biogenex Laboratories, Fremont CA) and FC receptor blocker (Innovex Biosciences, Richmond CA) for 20 min at RT. Primary antibodies (ULBP1–ULBP5) and MICA were applied at 5 μ g/ml overnight at 4 °C. Multilink secondary antibody (Biogenex Laboratories, Fremont CA) were applied for 30 min at RT, followed by streptavidin-labeled peroxidase for 30 min. For MICA staining, rabbit anti-mouse secondary antibody was used (Biogenex Labs, Fremont, CA) for 30 min following with Turbo DAB chromogen.

For microglial activation, mice were perfused and fixed as described [14]. Coronal sections that contained the hippocampus were cut at 40 μ m thickness and collected in tissue cryoprotectant solution in 96-well plates. Free floating sections were washed in PBS, incubated with antibodies against CD68 (Serotec, Oxford, UK), or Iba1 (Abcam, Cambridge, MA), phenotypic markers for microglia, and dual-stained with bromodeoxyuridine (BrdU) (Abcam). Colocalization of BrdU and the combination of various phenotypic markers in selected sections were evaluated using a confocal laser scanning microscope (Zeiss LSM700; Oberkochen, Germany).

Cell counting in the hippocampus was using a counting frame and a sampling grid of 75 μ m $\times$ 75 μ m at a magnification of 63 $\times$. The estimated number of the target cells was

calculated as the mean of the number of the target cells in the right and left dentate gyrus of the mouse brain. Statistically significant difference between control and irradiated mice was determined using *t* test.

Results

Expression of stress antigens in normal human astrocytes

NKG2D ligands are not typically detected on normal healthy cells, with the exception of highly proliferative cell types such as gastric and glandular epithelial cells [15, 16]. However, the potential expression of such ligands in stressed conditions on primary brain tissue creates the safety concerns of $\gamma\delta$ T cells therapy and dictates investigation of expression of NKG2D ligands in normal and radiation-stressed brain parenchyma cells. Cultured primary astrocytes serve as a broadly utilized system to evaluate in vitro tumor-parenchyma interaction [17, 18] and post-radiation consequences to the normal brain in rodent models [14, 19]. Here, we used human primary astrocytes as an easily available surrogate model of normal human brain parenchyma to evaluate potential side effects of $\gamma\delta$ T cell therapy. We examined surface expression of MICA, MICB and ULBP proteins (1, 2, 3, 4) on primary human astrocytes after 24 and 72 h of irradiation with two doses of irradiation (250 and 500 cGy). In addition, we evaluated an expression of stress markers on primary astrocytes treated with TMZ as the most relevant drug in glioma treatment. Moderate constitutive expression of MICA and slight ULBP2 expression were noted in untreated cultured astrocytes (Fig. 1a, left panel); however, irradiation did not result in up-regulation of NKG2D ligand expression. TMZ treatment also did not trigger up-regulation of any tested markers above basally expressed levels at 4 and 24 h post-treatment. TMZ-treated U251, as expected, demonstrated up-regulation of MICA, MICB and ULBP3 after TMZ treatment (Fig. 1b).

Cytotoxic effect of $\gamma\delta$ T cell on normal human astrocytes

Next, we examined whether expanded $\gamma\delta$ T cells are cytotoxic to normal or irradiation-stressed human astrocytes. Cytotoxicity against astrocytes was assessed 24 and 72 h post-irradiation applied at two doses (250 and 500 cGy). The astrocyte cytotoxicity profile was very similar for all conditions tested and at a starting effector-to-target ratio of 5:1 ranged from 6 to 11 % of dead cells (Fig. 2). The percentages of killed cells have moderately increased upon escalating the number of effector cells reaching 17–19 % at effector-to-target ratio of 20:1. Importantly, the same preparation of $\gamma\delta$ T cells demonstrated expected

dose-dependent toxicity toward the sensitive glioma cell line U251 with average of 53 % achieved at the highest effector-to-target (E:T) ratio tested. Depending on the potency of the $\gamma\delta$ T cell product following the culture preparation, cytotoxicity against U251 cells ranged from 35 to 65 % at a 20:1 ratio.

Three different primary $\gamma\delta$ T cell expansion cultures were used to evaluate cytotoxicity on primary astrocytes with or without irradiation. In all experiments, $\gamma\delta$ T cell cytotoxicity reached a plateau of approximately 12 % at 10:1 E:T ratio and was not progressive at higher dose of effector cells. Thus, in our experimental conditions, $\gamma\delta$ T cells demonstrated less cytotoxicity toward astrocytes than the control glioma cell line, which was statistically significant at ratios 10:1 and 20:1 ($p = 0.008$) and did not show the dose-dependent killing effect upon escalating the number of $\gamma\delta$ T cells.

Microglial activation and expression of stress ligand in irradiated mouse brain

In vivo testing of $\gamma\delta$ T cell therapy's safety requires an adequate disease-treatment model. Small rodent models (mice and rats) are commonly used to evaluate irradiation damage to the brain and investigate mechanisms of radiation injuries and toxicities.

The most appropriate model for our study would be presented by an immunocompetent mouse with established syngeneic brain tumors given consequent irradiation and injection of allogeneic mouse $\gamma\delta$ T cells. Some biological and technical issues, however, arise in using this model and translating results. Most pertinent are biological differences between mouse and human $\gamma\delta$ T cells and the inability to expand mouse $\gamma\delta$ T cells with bisphosphonates. As a surrogate solution, we have looked at the irradiated mouse brains in search for indicators of inflammation and expression of stress ligands in brain parenchyma.

Cells dual immunoreactive for BrdU and CD68 or Iba1 were used as a measure of activated microglia. In non-irradiated mouse brain, only occasional BrdU/CD68 or BrdU/Iba1-positive cells were seen. After 17 Gy, there was a generalized increase in microglia that demonstrated throughout the brain including the hippocampus (Fig. 3). To provide a quantitative assessment of microglial activation, a stereologic estimate of the number of BrdU-/CD68-positive or BrdU/Iba1-positive cells in the hippocampus was performed. A 17-Gy dose of radiation resulted in a significant increase in the number of activated microglia compared with non-irradiated controls (BrdU/CD68 cells, $p < 0.01$; BrdU/Iba1 cells, $p < 0.01$ compared with control).

The potential of radiation to induce delayed brain inflammation was evaluated at 70 days after irradiation. Brain sections were stained with CD3 AB to detect T cells for potential inflammatory response and MULT-1 AB to evaluate the

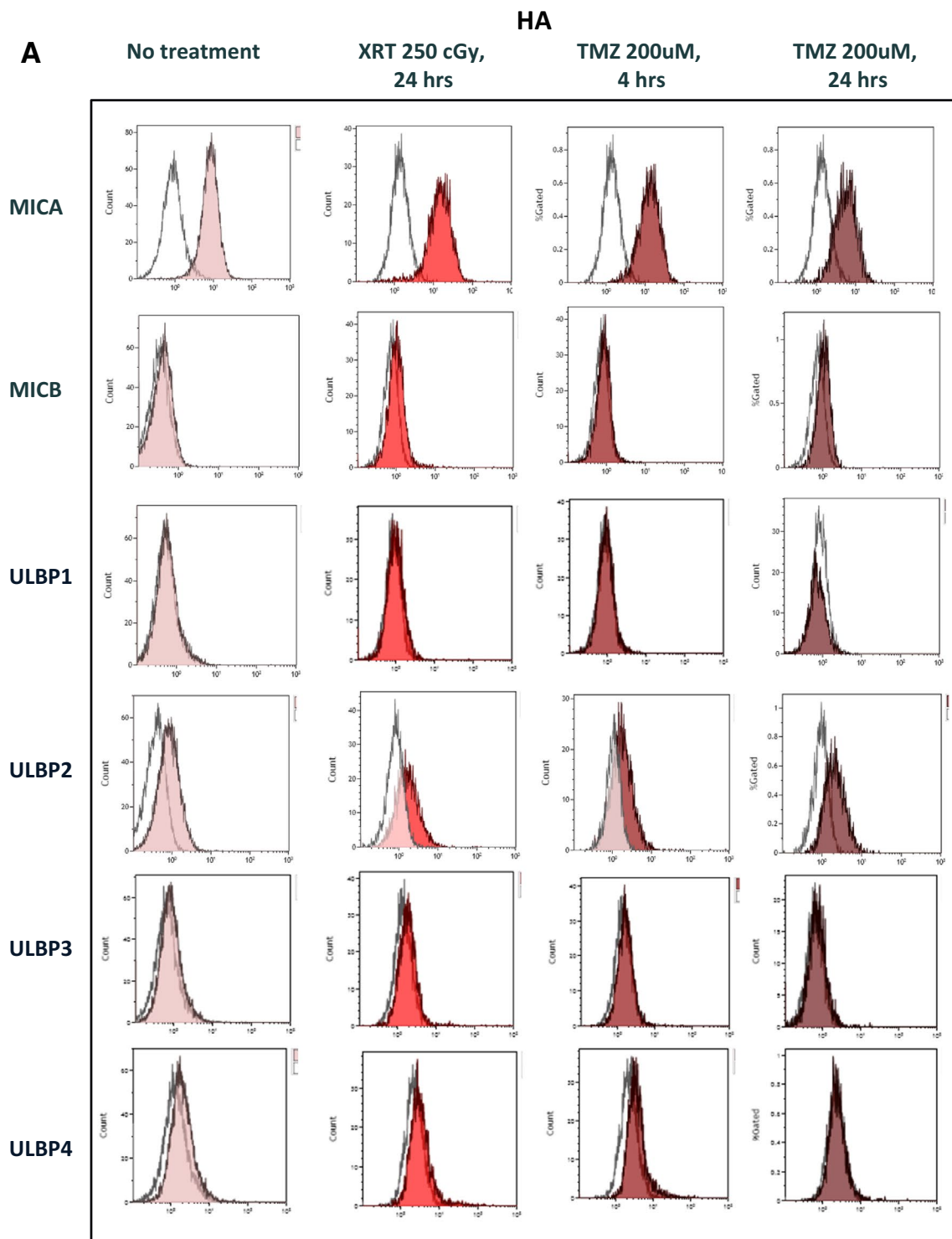


Fig. 1 Comparison of stress antigen expression on cultured human astrocytes (**a**) and U251 (**b**) without treatment, 24 h post-irradiation (250 cGy) and TMZ treatment (4 and 24 h post-treatment). The *no-shade histogram* stands for isotype control; the *gray-shaded his-*

tograms are the cells stained with a marker AB. Note constitutive expression of MICA by normal astrocytes. Only mild up-regulation of MICA expression on astrocytes is noted following irradiation, and no up-regulation was detected after TMZ treatment

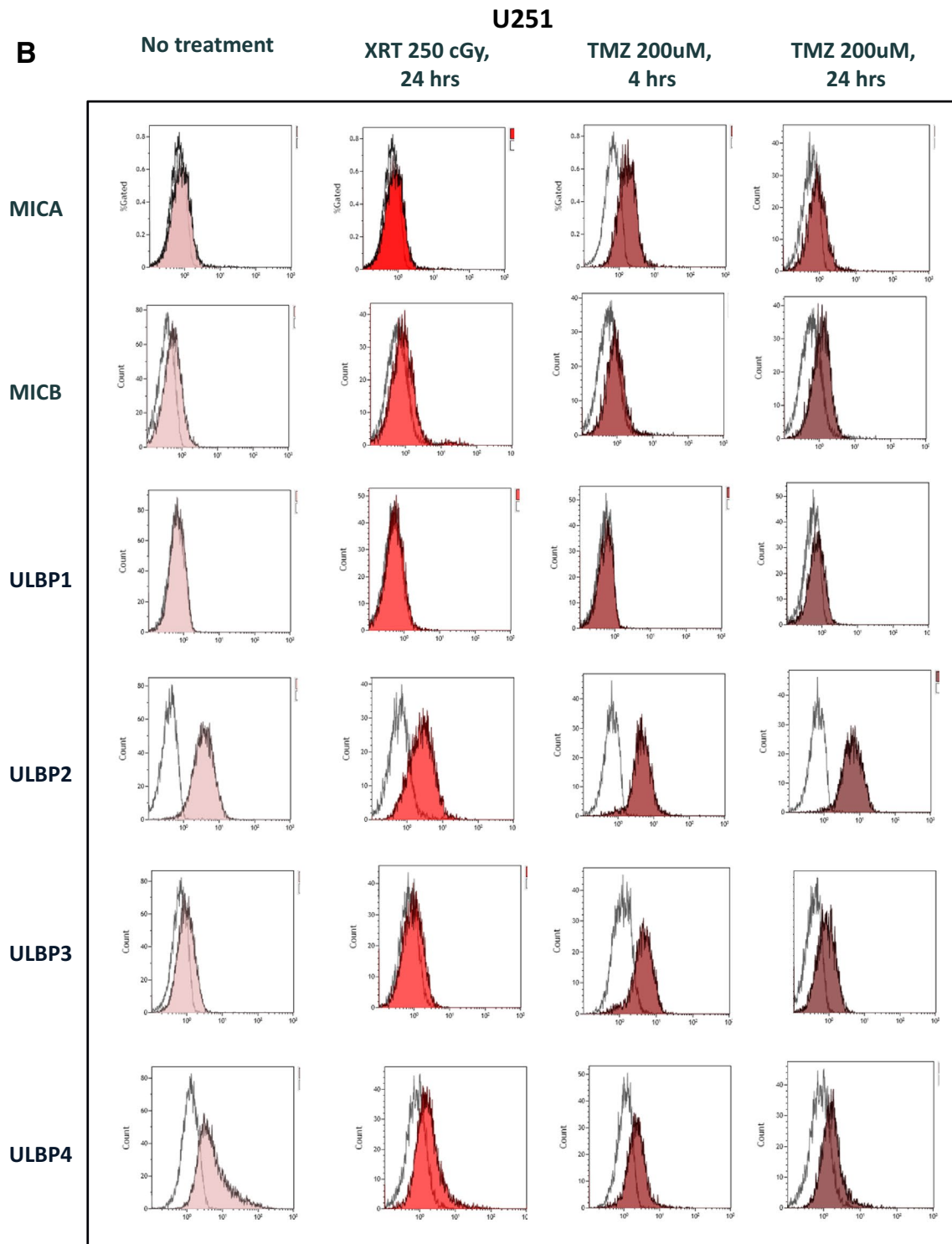


Fig. 1 continued

expression of mouse stress ligand in brain parenchyma. T cell infiltration did not differ between the control and the irradiated brain (Fig. 4). A low level of expression of MULT-1 was detected in both normal and irradiated mouse brain;

however, the intensity of staining and pattern of expression was the same in both control and irradiated groups.

Thus, despite evidences for neuroinflammation in the mouse brain at 70 days after irradiation, there

was no evidence for over-expression of the NKG2D ligand MULT-1 compared with non-irradiated mice brain.

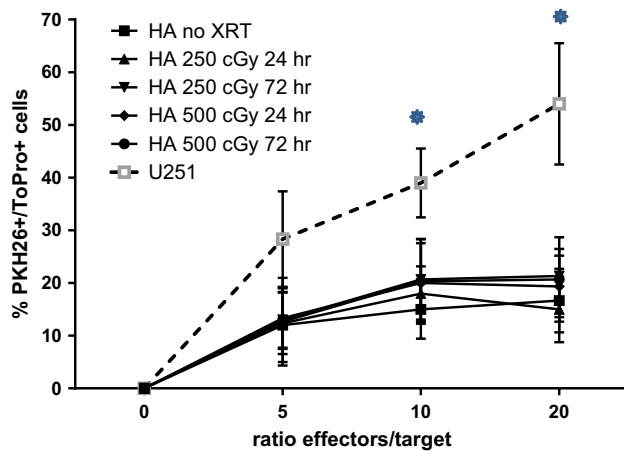
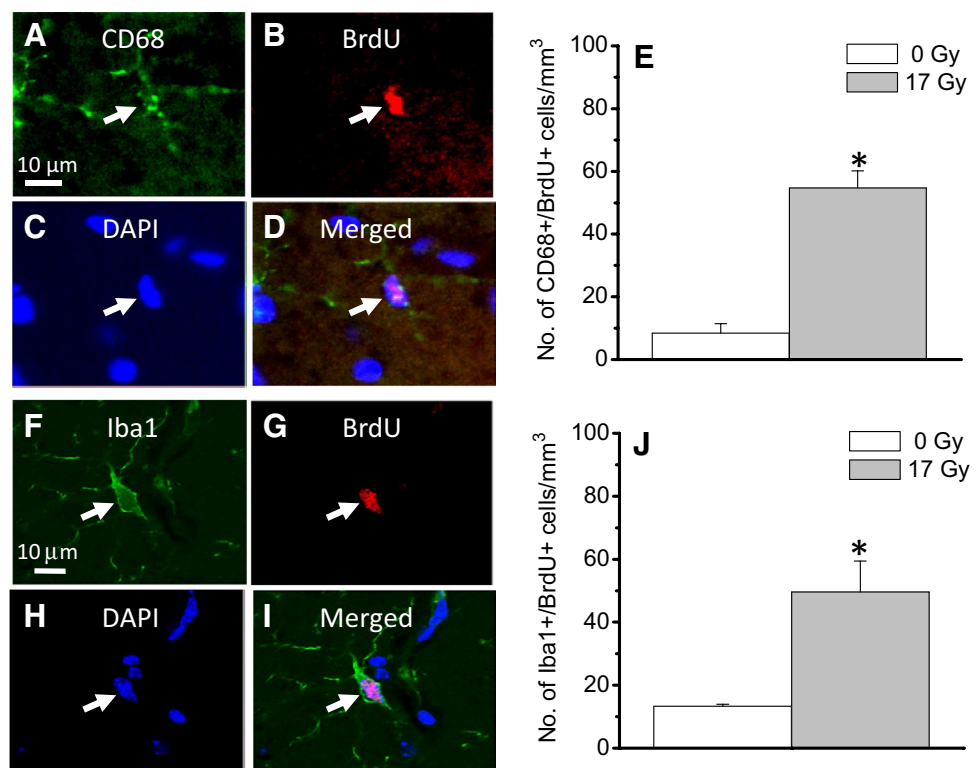


Fig. 2 Cytotoxicity of $\gamma\delta$ T cell to normal human astrocytes. Ex vivo expanded/activated $\gamma\delta$ T cells were cultured in increasing E:T ratios with tumor glioma cell line (U251) and primary human astrocytes (non-irradiated and irradiated). Cytotoxic assay was performed 24 and 72 h post-irradiation. Incremental cytotoxic activity of $\gamma\delta$ T cells with target tumor cell line is noted, while astrocytes demonstrated lower killing efficiency regardless of irradiation and lack of dose escalating toxicity. Data were presented as mean percentage of target cell lysis ($\pm$ SD) obtained in three independent experiments. The difference between cytotoxicity values for U251 and primary astrocytes were statistically significant at ratios 10:1 and 20:1 ($p = 0.008$)

Fig. 3 Microglia activation in irradiated mouse hippocampus. An activated microglial cell is CD68 positive (**a**, green, arrow) and demonstrates BrdU incorporation (**b**, BrdU, red; **c**, DAPI; **d**, merged). An Iba1-positive microglial cell (**f**, green, arrow) shows BrdU incorporation (**g**, BrdU; **h**, DAPI; **i**, merged). There is an increase in the number of CD68-/BrdU-positive (**e**) and Iba1-/BrdU-positive cells (**j**) in mouse hippocampus at 70 days after 17 Gy (* $p < 0.01$, compared with non-irradiated controls, t test). Error bars are SEM. $n = 3$ mice per group



Expression of stress ligands in GBM and breast tumor metastasis in the human brain

We have also evaluated the expression of NKG2D ligands in available human clinical material. It was shown that the expression levels of NKG2D ligands on tumor cells correlates with efficacy of both $\gamma\delta$ and NK cell-mediated killing [20]. Thus, we investigated the expression of these ligands in brain tumors and normal brain parenchyma as correlative markers of efficacy and safety of $\gamma\delta$ T cell injections to the patients with brain tumors. Sections of the normal human brain and brain with metastatic breast cancer tumors and GBM multiforme were stained with antibodies to MICA and ULBP proteins (1, 2, 3 and 4). Breast cancer tumor cells and GBM cells demonstrated a prominent expression of MICA and ULBP1 and 2 (Fig. 5). Staining for ULBP3 and 4 in both tumors was not as strong, but still positive compared with the normal brain parenchyma. Breast cancer metastasis to the brain usually appeared as localized nodules sequestered from normal brain parenchyma, while the invasive GBM cells spread through brain parenchyma. This is clearly seen in GBM sections stained with ULBP2 and 4. In summary, the constitutive level of expression of NKG2D ligands on tumor cells was qualitatively higher compared with brain parenchyma, providing a clear rationale to use cytotoxic NKG2D axis of $\gamma\delta$ T cells for selective tumor killing.

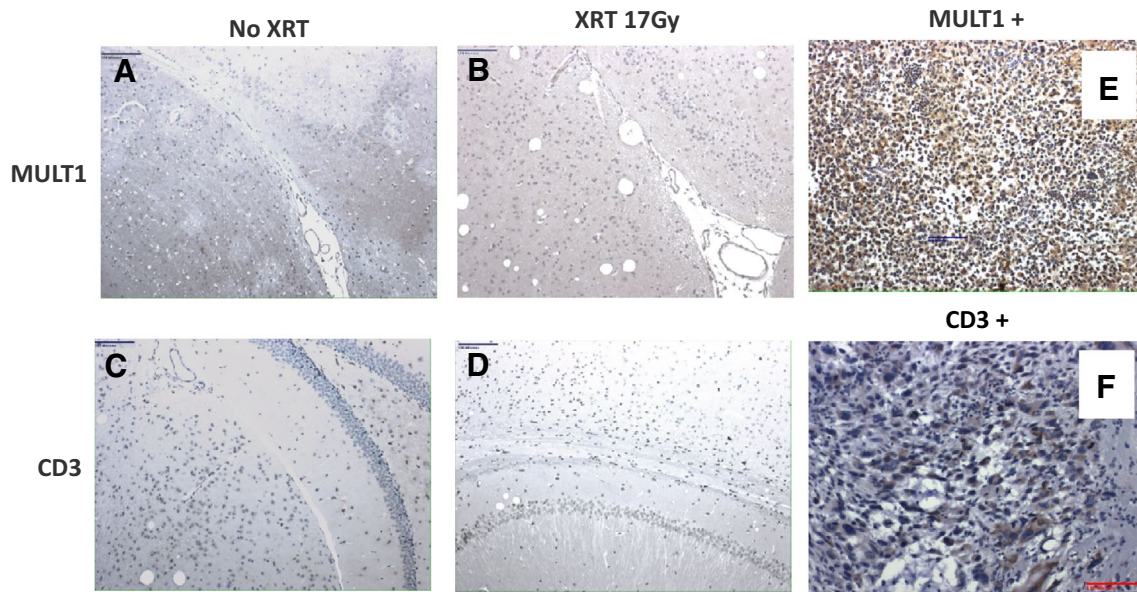


Fig. 4 Lack of up-regulation of stress-associated antigens and CD3 infiltration in irradiated mice brain. The whole brain of WT C56BL/6 mice (**b**, **d**) was radiated using an anterior–posterior and posterior–anterior pair of beams of 160 kV X-rays at a dose of 17 Gy. Control mice (**a**, **c**) received no radiation. Mice were killed and brains harvested at 70 days. **a** and **b** compare stress antigen MULT-1 staining between control and irradiated mice. **c** and **d** compare staining

with anti-CD3. Irradiated mice brain sections are both negative for MULT-1 and anti-CD3, suggesting that stress antigen expression is not up-regulated and lymphocyte infiltration is not present. Positive controls for MULT-1 and CD3 staining are shown on **e** and **f** correspondingly and presented by GL261 tumors established in Bl6 mouse brain. All pictures were taken at $\times 10$ magnification; the *scale bars* in the *upper left corner* correspond to 100 microns

Discussion

Cancer immunotherapy was rated by the Science editorial board as a Breakthrough of the Year 2013 [21] and acknowledged as a development of an evolving new paradigm in the treatment of cancer. In this study, we sought to describe the potential risks that will be encountered in the clinical setting if innate lymphocyte therapy is applied as treatment for high-grade gliomas. Specifically, we set up to investigate the parameters indicative of safety of ex vivo expanded $\gamma\delta$ T cells introduced in the context of a post-irradiated brain. This issue is important to predict the safety of $\gamma\delta$ T cell-based immunotherapy applied in combination with toxic regimens that are currently standard of care in brain tumors. This is also a necessary characterization of the cell product before clinical trials.

The safety issues of immune cell therapies can relate to cell infusion in general, such as the unwanted migration of injected cells and the rejection of allogeneic cells in the immune host or can be cell type specific. For instance, a phenomenon known as the cytokine-release syndrome that includes high-grade cyclical fevers and hypotension has been reported after infusion of CAR T cells and is a result of overproduction of pro-inflammatory cytokines upon T cell triggering [22]. The overall safety of $\gamma\delta$ T cell therapy in context of various somatic solid tumors was tested in

several clinical trials [23] and documented a lack of severe side effects along with the presence of common side effects (flu-like symptoms, fever and post-injection thrombophlebitis) [24]. In these protocols, cells were given by IV injections (systemically). Our strategy proposed for glioma treatment capitalizes on intracranial $\gamma\delta$ T cell injections. Such treatment modality, along with the benefits of avoiding systemic toxicities, can produce toxicities not revealed in previous studies. While direct cytotoxicity of expanded and activated $\gamma\delta$ T cells to normal brain cell seems unlikely, and our previous experiments have demonstrated no in vitro toxicity to normal astrocytes [13, 25], an attempt to boost efficiency of $\gamma\delta$ T cell therapy by scheduling it after toxic pre-treatments (irradiation or chemotherapy or both) requires introduction of $\gamma\delta$ T cells into a potentially inflamed or stressed environment, where innate effector cells may exert their cytotoxic function to the stressed brain parenchyma. To exclude such possibilities, we have looked at several parameters indicative of $\gamma\delta$ T safety in models that are available and easily accessible.

The NKG2D axis is a recognition system that is increasingly being used in cell-based cancer immunotherapy utilizing constitutive expression of NKG2D ligands on tumors cell lines and primary tumors of different origin [15, 26]. We have previously confirmed the expression of NKG2D ligands on major glioma cell lines and primary tumor

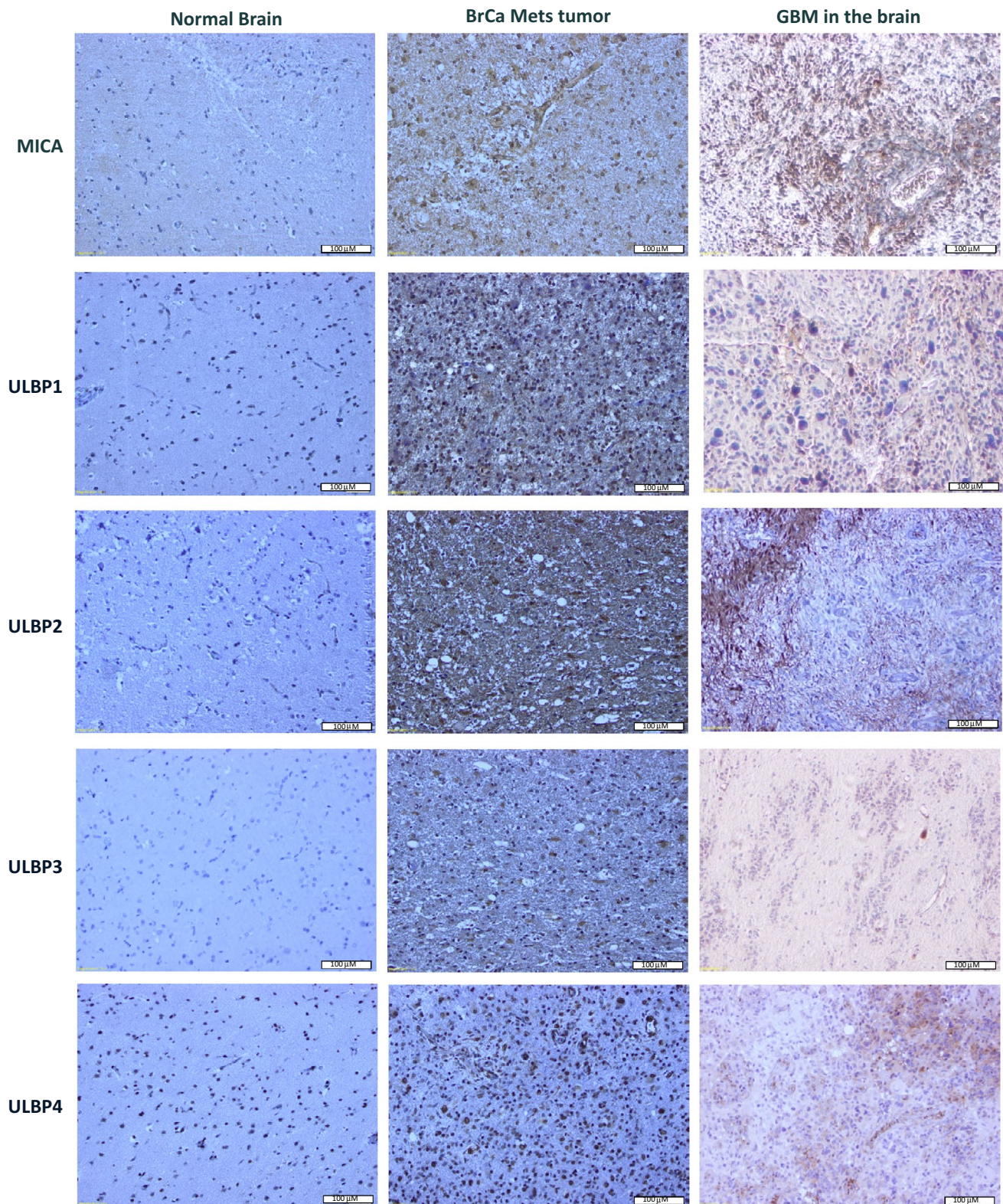


Fig. 5 Stress-antigen expression from metastatic breast tumor and adjacent normal brain. Note strong expression of ULBP1 and ULBP2 and mild expression of and ULBP3 ULBP4 on metastatic breast tumors (*middle panels*) with no stress antigen expression on normal brain (*left panels*). *Right panels* present GBM samples. ULBP2 and

ULBP4 staining clearly demonstrate differences in NKG2D ligand expression on tumor cells and normal brain parenchyma. All pictures were taken at $\times 10$ magnification; the *scale bars* in the *lower right corner* correspond to 100 microns

xenografts [13]. In this study, we have strengthened these observations by confirming NKG2D ligand over-expression on human primary brain tumors by IHC staining of available clinical material.

Even further up-regulation of NKG2D ligands on tumor cells is noted as an effect of chemotherapy as a general cell stressor. We have demonstrated transient up-regulation of NKG2D ligands (MICA/B, ULBP1,2,4) after TMZ exposure in TMZ-resistant cell line [20]. This phenomenon was extensively investigated in various tumor cell lines [27], primary tumors [28], animal models and in patients [29, 30]. In these studies, up-regulation of ligands expression was the most common observation, although the pattern and the level of increase were variable and not predictive [30]. Levels of expression of NKG2D ligands in post-irradiated tumors are much less investigated. Yet, some studies documented up-regulation of these ligands after XRT exposure similarly to the effect of chemotherapy [31–34]. These data confirmed strong rationale to use $\gamma\delta$ T cells expressing NKG2D receptors as effectors to kill GBM tumors in pre-clinical and clinical settings.

NKG2D ligands are generally not expressed on quiescent healthy tissues [16, 26]. Although mRNA transcripts are detected in normal healthy cells, such as ULBP transcripts, proteins often were not detected by immunohistochemistry. One exception is highly proliferative tissues or cells such as bone marrow stem cells or cultured human T cells stimulated with mitogens. We have shown minimal to moderate constitutive expression of MICA and ULBP2 in untreated cultured astrocytes. Although we cannot rule out the culture itself as a source of stress, examination of normal brain tissue suggests that NKG2D ligand expression is undetectable in vivo (Fig. 5). The lack of expression on normal healthy cells provides one potential safety concern for immunotherapy with NKG2D-expressing effectors. Much less data are available documenting the effect of irradiation on expression of NKG2D ligands in normal non-malignant cells. In the study of Riederer et al. [35], primary human umbilical vein endothelial cells irradiated with a single dose of 4 Gy did not change the expression levels of MICA, MICB or ULBPs. At the same time, it is well known that an activation of DNA damage response (DDR) pathway, as a part of common radiation response, induces expression of NKG2D ligands in mouse and human cells [36]. The level of response, consequences and expressions of stress ligands in post-irradiated non-malignant cells most likely vary as it generally had been noticed for tumor cell lines and primary tumors. Besides being dose dependent, such responses are affected by cell type, metabolic state and genetic makeup. It is likely also that expression of NKG2D ligands in tumor and healthy cells differs even under the same stress conditions and thus needs to be investigated for each experimental setting.

Importantly, irradiation in our experiments did not result in up-regulation of expression of NKG2D ligands in normal human astrocytes. We did not detect significant expression of NKG2D ligands in human brain parenchyma while examining IHC stained paraffin sections of human brain tumors (Fig. 5). These two lines of data suggest important safety prerequisites for $\gamma\delta$ T cell infusions applied to brain tumor treatment.

Concurrent with low NKG2D ligands expression, the cytotoxicity of $\gamma\delta$ T cells toward irradiated astrocytes was significantly less and did not show a dose-dependent pattern compared with the U251 glioma cell line. Some background cytotoxicity toward irradiated astrocytes was apparently related to the process of elimination of dead and/or apoptotic cells. Together with NK cells, $\gamma\delta$ T cells are known to participate in lymphoid stress-surveillance response, which is increasingly implicated in tumor immunology, autoimmunity, infectious disease and inflammation. Theoretic background for stress surveillance is established [37]; for instance, $\gamma\delta$ T cells have been shown to be important in the early inflammatory response to burn injury [38]; early activation is detected in trauma severe systemic inflammatory response syndrome [39]. However, data on direct cytotoxicity (especially on quantitative level) of activated $\gamma\delta$ T cells against non-malignant or uninfected stressed cells are very limited.

In general, irradiation, especially with higher doses (e.g., doses ≥ 2 Gy), exerts pro-inflammatory effects and results in inflammatory processes as common toxicity of radiation therapy [40]. One well-documented consequence of brain irradiation is neuroinflammation that is manifested as an acute phase response as modulation of pro-inflammatory cytokine expression, such as TNF α , IL-1 and IL-6 [8]. Evidence of late chronic phase was also experimentally confirmed on rodent models and includes elevation of inflammatory cytokines in mouse brain for up to 6 months [41], increased expression of the CCR2 [42] and adhesion molecules (ICAM-1) [14], and persistent microglial and astrocyte activation [43, 44]. A detailed characterization of the neuroinflammatory response in the C57BL/6 mouse model after single-dose cranial irradiation was performed in the study of Moravan et al. [45]. It demonstrated a bi- or multiphasic inflammatory response that started with an acute recruitment of neutrophils and rapid endothelial and glial activation followed by a delayed infiltration of T cells beginning at 1 month post-irradiation and accompanied by the appearance of the MHC II-positive dendritic cells. These post-irradiation conditions may potentially trigger a cytotoxic response of $\gamma\delta$ T cells introduced in post-irradiated brain.

Although we documented microglia activation as in the mouse brain at 70 days after irradiation (Fig. 3), there were no evidences of immune CD3 + infiltrations after

irradiation with either 10 or 17 Gy. An expression of mouse NKG2D ligand, MULT-1, in the mouse brain after irradiation was also found unchanged compared with controls at this late stage. This suggests that the stressful effect of irradiation subsides at least in regard to up-regulation of NKG2D ligands, which dismisses one potential safety concern for immunotherapy with NKG2D-expressing effectors.

Conclusion

In summary, while general clinical safety of $\gamma\delta$ T cell therapy as adjunct therapy for solid tumors was evaluated and confirmed in available to date clinical trials, infusion of activated immune cells creates a concern for pathologic immune reaction and effects. Our study indicates that several parameters selected as safety indicators of $\gamma\delta$ T cell therapy have complied with the hypothesis that $\gamma\delta$ T cell therapy can be well tolerated in combination treatment settings. Some practical issues still need addressing, including an investigation of optimal scheduling and dosage for irradiation and immunotherapy. Most likely, there will be requirements for a therapeutic window of $\gamma\delta$ T cell therapy and the optimal dose of radiation treatment to ensure maximal synergy of treatment combination in tumoricidal effect and minimal impact on brain parenchyma. Further studies are needed to define those optimal parameters and test the safety of $\gamma\delta$ T cells in clinical settings for treatment of aggressive brain tumors.

Acknowledgments We thank Dr. Mitchell S. Berger, Department of Neurological Surgery, University of California, San Francisco for making available FFPE samples of human brain tumors. Support from Elsa U Pardee Foundation (Lawrence S. Lamb) is acknowledged and appreciated.

Conflict of interest The authors declare that they have no conflict of interest.

References

- Ishikawa E, Takano S, Ohno T, Tsuboi K (2012) Adoptive cell transfer therapy for malignant gliomas. *Adv Exp Med Biol* 746:109–120
- Bielamowicz K, Khawja S, Ahmed N (2013) Adoptive cell therapies for glioblastoma. *Front Oncol* 3:275
- Lamb LS (2009) Gammadelta T cells as immune effectors against high-grade gliomas. *Immunol Res* 45:85–95
- Vantourout P, Hayday A (2013) Six-of-the-best: unique contributions of $\gamma\delta$ T cells to immunology. *Nat Rev Immunol* 13:88–100
- Beetz S, Marischen L, Kabelitz D, Wesch D (2007) Human gamma delta T cells: candidates for the development of immunotherapeutic strategies. *Immunol Res* 37:97–111
- Kabelitz D, Wesch D, He W (2007) Perspectives of gammadelta T cells in tumor immunology. *Cancer Res* 67:5–8
- Preusser M, de Ribaupierre S, Wöhrer A, Erridge SC, Hegi M et al (2011) Current concepts and management of glioblastoma. *Ann Neurol* 70:9–21
- Mukherjee D, Coates PJ, Lorimore SA, Wright EG (2014) Responses to ionizing radiation mediated by inflammatory mechanisms. *J Pathol* 232:289–299
- Zhao W, Robbins ME (2009) Inflammation and chronic oxidative stress in radiation-induced late normal tissue injury: therapeutic implications. *Curr Med Chem* 16:130–143
- Friedman EJ (2002) Immune modulation by ionizing radiation and its implications for cancer immunotherapy. *Curr Pharm Des* 8:1765–1780
- Rödel F, Frey B, Gaip U, Keilholz L, Fournier C et al (2012) Modulation of inflammatory immune reactions by low-dose ionizing radiation: molecular mechanisms and clinical application. *Curr Med Chem* 19:1741–1750
- Rödel F, Frey B, Multhoff G, Gaip U (2015) Contribution of the immune system to bystander and non-targeted effects of ionizing radiation. *Cancer Lett* 356:105–113
- Bryant NL, Gillespie GY, Lopez RD, Markert JM, Cloud GA et al (2011) Preclinical evaluation of ex vivo expanded/activated $\gamma\delta$ T cells for immunotherapy of glioblastoma multiforme. *J Neurooncol* 101:179–188
- Wu KL, Tu B, Li YQ, Wong CS (2010) Role of intercellular adhesion molecule-1 in radiation-induced brain injury. *Int J Radiat Oncol Biol Phys* 76:220–228
- Spear P, Wu MR, Sentman ML, Sentman CL (2013) NKG2D ligands as therapeutic targets. *Cancer Immun* 13:8
- Raulet DH, Gasser S, Gowen BG, Deng W, Jung H (2013) Regulation of ligands for the NKG2D activating receptor. *Annu Rev Immunol* 31:413–441
- Kim SJ, Kim JS, Park ES, Lee JS, Lin Q et al (2011) Astrocytes upregulate survival genes in tumor cells and induce protection from chemotherapy. *Neoplasia* 13:286–298
- Wang L, Cossette SM, Rarick KR, Gershan J, Dwinell MB et al (2013) Astrocytes directly influence tumor cell invasion and metastasis in vivo. *PLoS One* 8:e80933
- Moore ED, Kooshki M, Metheny-Barlow LJ, Gallagher PE, Robbins ME (2013) Angiotensin-(1–7) prevents radiation-induced inflammation in rat primary astrocytes through regulation of MAP kinase signaling. *Free Radic Biol Med* 65:1060–1068
- Lamb LS, Bowersock J, Dasgupta A, Gillespie GY, Su Y et al (2013) Engineered drug resistant $\gamma\delta$ T cells kill glioblastoma cell lines during a chemotherapy challenge: a strategy for combining chemo- and immunotherapy. *PLoS One* 8:e51805
- Couzin-Frankel J (2013) Breakthrough of the year 2013. *Cancer immunotherapy. Science* 342:1432–1433
- Morgan RA, Yang JC, Kitano M, Dudley ME, Laurencot CM et al (2010) Case report of a serious adverse event following the administration of T cells transduced with a chimeric antigen receptor recognizing ERBB2. *Mol Ther* 18:843–851
- Nicol AJ, Tokuyama H, Mattarollo SR, Hagi T, Suzuki K et al (2011) Clinical evaluation of autologous gamma delta T cell-based immunotherapy for metastatic solid tumours. *Br J Cancer* 105:778–786
- Gomes AQ, Martins DS, Silva-Santos B (2010) Targeting $\gamma\delta$ T lymphocytes for cancer immunotherapy: from novel mechanistic insight to clinical application. *Cancer Res* 70:10024–10027
- Bryant NL, Suarez-Cuervo C, Gillespie GY, Markert JM, Nabors LB et al (2009) Characterization and immunotherapeutic potential of gammadelta T-cells in patients with glioblastoma. *Neuro Oncol* 11:357–367
- Coudert JD, Held W (2006) The role of the NKG2D receptor for tumor immunity. *Semin Cancer Biol* 16:333–343
- Schmudde M, Braun A, Pende D, Sonnemann J, Klier U et al (2008) Histone deacetylase inhibitors sensitize tumour cells for cytotoxic effects of natural killer cells. *Cancer Lett* 272:110–121
- Rohner A, Langenkamp U, Siegler U, Kalberer CP, Wodnar-Filipowicz A (2007) Differentiation-promoting drugs up-regulate

- NKG2D ligand expression and enhance the susceptibility of acute myeloid leukemia cells to natural killer cell-mediated lysis. *Leuk Res* 31:1393–1402
29. Poggi A, Catellani S, Garuti A, Pierri I, Gobbi M et al (2009) Effective in vivo induction of NKG2D ligands in acute myeloid leukaemias by all-trans-retinoic acid or sodium valproate. *Leukemia* 23:641–648
 30. Rosental B, Appel MY, Yossef R, Hadad U, Brusilovsky M et al (2012) The effect of chemotherapy/radiotherapy on cancerous pattern recognition by NK cells. *Curr Med Chem* 19:1780–1791
 31. Kim JY, Son YO, Park SW, Bae JH, Chung JS et al (2006) Increase of NKG2D ligands and sensitivity to NK cell-mediated cytotoxicity of tumor cells by heat shock and ionizing radiation. *Exp Mol Med* 38:474–484
 32. Bedel R, Thiery-Vuillemin A, Grandclement C, Balland J, Remy-Martin JP et al (2011) Novel role for STAT3 in transcriptional regulation of NK immune cell targeting receptor MICA on cancer cells. *Cancer Res* 71:1615–1626
 33. Xu X, Rao GS, Groh V, Spies T, Gattuso P et al (2011) Major histocompatibility complex class I-related chain A/B (MICA/B) expression in tumor tissue and serum of pancreatic cancer: role of uric acid accumulation in gemcitabine-induced MICA/B expression. *BMC Cancer* 11:194
 34. Butler JE, Moore MB, Presnell SR, Chan HW, Chalupny NJ et al (2009) Proteasome regulation of ULBP1 transcription. *J Immunol* 182:6600–6609
 35. Riederer I, Sievert W, Eissner G, Molls M, Multhoff G (2010) Irradiation-induced up-regulation of HLA-E on macrovascular endothelial cells confers protection against killing by activated natural killer cells. *PLoS One* 5:e15339
 36. Gasser S, Orsulic S, Brown EJ, Raulet DH (2005) The DNA damage pathway regulates innate immune system ligands of the NKG2D receptor. *Nature* 436:1186–1190
 37. Hayday AC (2009) Gammadelta T cells and the lymphoid stress-surveillance response. *Immunity* 31:184–196
 38. Schwacha MG (2009) Gammadelta T-cells: potential regulators of the post-burn inflammatory response. *Burns* 35:318–326
 39. Matsushima A, Ogura H, Fujita K, Koh T, Tanaka H et al (2004) Early activation of gammadelta T lymphocytes in patients with severe systemic inflammatory response syndrome. *Shock* 22:11–15
 40. Williams J, Chen Y, Rubin P, Finkelstein J, Okunieff P (2003) The biological basis of a comprehensive grading system for the adverse effects of cancer treatment. *Semin Radiat Oncol* 13:182–188
 41. Lee WH, Sonntag WE, Mitschelen M, Yan H, Lee YW (2010) Irradiation induces regionally specific alterations in pro-inflammatory environments in rat brain. *Int J Radiat Biol* 86:132–144
 42. Rola R, Sarkissian V, Obenaus A, Nelson GA, Otsuka S et al (2005) High-LET radiation induces inflammation and persistent changes in markers of hippocampal neurogenesis. *Radiat Res* 164:556–560
 43. Schindler MK, Forbes ME, Robbins ME, Riddle DR (2008) Aging-dependent changes in the radiation response of the adult rat brain. *Int J Radiat Oncol Biol Phys* 70:826–834
 44. Greene-Schloesser D, Moore E, Robbins ME (2013) Molecular pathways: radiation-induced cognitive impairment. *Clin Cancer Res* 19:2294–2300
 45. Moravan MJ, Olschowka JA, Williams JP, O'Banion MK (2011) Cranial irradiation leads to acute and persistent neuroinflammation with delayed increases in T-cell infiltration and CD11c expression in C57BL/6 mouse brain. *Radiat Res* 176:459–473